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## **SECB3203 PROGRAMMING FOR BIOINFORMATICS**

### **Report**

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Section 01 Group 12

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Date: 2026

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1.0 INTRODUCTION

Leukemia is a heterogeneous group of hematologic malignancies characterized by the rapid production of abnormal white blood cells (leukocytes). It comprises **four major types**: Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Chronic Lymphocytic Leukemia (CLL), and Chronic Myeloid Leukemia (CML). These types are differentiated by the lineage of the affected cells (myeloid vs. lymphoid) and the rate of disease progression (acute vs. chronic).

|           | Acute                                     | Chronic                                     |
|-----------|-------------------------------------------|---------------------------------------------|
| Myeloid   | <b>AML</b><br>Acute<br>myeloid leukemia   | <b>CML</b><br>Chronic<br>myeloid leukemia   |
| Lymphatic | <b>ALL</b><br>Acute<br>lymphatic leukemia | <b>CLL</b><br>Chronic<br>lymphatic leukemia |

Figure 1: Leukemia and Lymphoma: The Differences and Similarities  
([rocky-mountain-cancer-center](#))

Leukemia is considered critical primarily because it is a fast-acting cancer of the blood-producing system. It quickly fills the bone marrow with dysfunctional cells, preventing the production of healthy blood components like red blood cells and platelets. The prognosis varies significantly by type; for instance, Acute Myeloid Leukemia (AML) has a 31.9% five-year survival rate (Cooke & Taylor, 2025). Therefore, rapid and accurate diagnosis is essential for immediate treatment intervention.

## 1.1 Problem Background

Common **clinical diagnosis methods** include Complete Blood Count (CBC), peripheral blood smear, bone marrow biopsy, and flow cytometry. Among these, the bone marrow biopsy is considered the "gold standard" with near 100% accuracy, as it allows for direct examination of cellular morphology and blast percentage.

However, these traditional methods face significant limitations such as **invasiveness**, as the bone marrow biopsies often involve painful procedures. These approaches are also **time-consuming** as a full cytogenetic analysis can take up to 14 days. While methods like blood smears rely heavily on the **expertise** of the pathologist, which can lead to human error in ambiguous cases.

To address these limitations, **bioinformatics approaches** are being integrated into the cancer study. The use of gene expression profiling for leukemia diagnosis was pioneered by Golub et al. (1999), who demonstrated that cancer classification could be achieved solely through gene expression monitoring without relying on morphological approach. This field was further advanced by the **MILE Study** (Microarray Innovations in LEukemia), which aimed to standardize diagnosis using microarray data across international laboratories (Haferlach et al., 2010).

## 1.2 Problem Statement

While traditional clinical methods are accurate, they are invasive and involved high dependency on expertise. Furthermore, undifferentiated leukemias can present ambiguous morphological features, making accurate diagnosis difficult. There is a need for a **unified diagnostic tool** that utilizes high-dimensional biological data. By analyzing the expression levels of specific genes, it is possible to come up with a diagnosis that is purely data-driven and potentially faster than common clinical approaches.

## 1.3 Objectives

The main objective of this project is to develop a bioinformatics-based machine learning model to accurately diagnose the four major leukemia types. The objectives are:

- To implement a data wrangling pipeline that extracts and processes gene expression data from a large dataset.
- To identify and map specific genetic biomarkers that distinguish between ALL, AML, CLL, and CML.
- To implement several different models for leukemia types classification.
- To find the best model which offers the highest accuracy and least error.

#### 1.4 Scopes

This project is defined by the following inclusions and exclusions:

|                       | <b>Inclusion (What the project will do)</b>                                                | <b>Exclusion (What the project will not do)</b>                                       |
|-----------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Data Source</b>    | Utilize secondary, publicly available high-throughput gene expression data from databases. | Generate or process primary sequencing data; rely on private or proprietary datasets. |
| <b>Classification</b> | Focus on classifying the four major leukemia types: ALL, AML, CLL, and CML.                | Attempt to classify rare types or predict patient survival/prognosis.                 |
| <b>Tools</b>          | Develop the analysis pipeline entirely in Python using standard bioinformatics libraries.  | Use specialized statistical software or build a full-scale web application/API.       |

*Table 1: Inclusions and exclusions of this project in different aspects*

## 2.0 DATA COLLECTION AND PRE-PROCESSING

### 2.1 Importing Dataset

The dataset was sourced from the **Gene Expression Omnibus (GEO)** under accession number **GSE13164** which is a project in Microarray Innovations in LEukemia (MILE) study Stage 2. The `GEOparse` library was used to download and parse the soft-formatted family file, which contains both sample metadata (GSM) and platform annotation tables (GPL7473).

| Characteristic                      | Detail                                                                          | Rationale for Selection                                                                                                |
|-------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Accession ID</b>                 | GSE13164 (MILE Study Stage 2)                                                   | Selected for its large sample size while maintaining reasonable download and processing feasibility on local hardware. |
| <b>Sample Size (<math>N</math>)</b> | $N = 1,152$ samples                                                             | Provides a robust cohort size to minimize overfitting and maximize statistical power for model evaluation.             |
| <b>Data Type</b>                    | Gene expression profiling by array (Roche AmpliChip Leukemia Custom Microarray) | Standardized microarray format, ideal for the planned Python analysis pipeline.                                        |
| <b>Leukemia Types</b>               | Includes ALL, AML, CLL, and CML                                                 | Direct support for the project's multi-class classification problem statement.                                         |

*Table 2: Dataset details and rationales of selection*

## 2.2 Data Wrangling

A comprehensive cleaning pipeline was implemented to transform the raw microarray data:

- **Sample Filtering:** Metadata was scanned using `regex` to identify and retain only samples labeled as ALL, AML, CLL, or CML. The details of each leukemia types were discarded, as well as ambiguous data.
- **Probe-to-Gene Mapping:** Raw probe IDs (`ID_REF`) were mapped to gene accessions (`GB_ACC`) using the platform annotation table. Probes with missing or "---" gene identifiers were removed.
- **Aggregation:** Since multiple probes can represent a single gene, expression values were grouped by gene accession and averaged to create a unique feature for each of the 1,420 genes.
- **Label Encoding:** Categorical labels (ALL, AML, CLL, CML) were converted into numeric codes (0–3) using `LabelEncoder`.

The output of the cleaned data is saved in 2 .csv files, the `GSE13164_cleaned_features.csv` contains the feature matrix (samples  $\times$  genes) and the `GSE13164_cleaned_labels.csv` contains the label matrix (samples  $\times$  labels). The shape of the 2 files are as follows:

|                                            | Role in Analysis | Final Shape (Samples x Features) |
|--------------------------------------------|------------------|----------------------------------|
| <code>GSE13164_cleaned_features.csv</code> | Features (X)     | (973, 1420)                      |
| <code>GSE13164_cleaned_labels.csv</code>   | Labels (Y)       | (973, 3)                         |

*Table 3: Files generated and their usage in the next phase*

## 2.3 Software and Hardware Requirements

The language used in this project is `Python 3.13.5` and Jupyter Notebook is used for better comprehension. While the libraries used are as follows:

| Library/Module                           | Purpose                                                                                |
|------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Data Handling &amp; Processing</b>    |                                                                                        |
| GEOparse                                 | Download and parse gene expression data from the GEO database.                         |
| pandas                                   | Create and manipulate dataframes to handle, clean, and structure the dataset.          |
| numpy                                    | For numerical operations, array manipulation, and mathematical functions.              |
| os                                       | Interact with the operating system, such as creating directories for saving data.      |
| re                                       | For regular expression operations to extract leukemia types from sample metadata text. |
| functools.reduce                         | Sequentially merge multiple sample dataframes into a single expression matrix.         |
| <b>Machine Learning &amp; Statistics</b> |                                                                                        |
| sklearn                                  | For data splitting, scaling, building regression models, and evaluation.               |
| scipy.stats                              | Perform statistical tests, specifically the one-way ANOVA, to find significant genes.  |
| tensorflow.keras                         | Build, train, and evaluate the deep neural network model for classification.           |
| <b>Data Visualization</b>                |                                                                                        |
| matplotlib                               | Create static, animated, and interactive visualizations and plots.                     |
| seaborn                                  | Create more attractive and informative statistical graphics, such as heatmaps.         |

*Table 4: Python libraries used in entire project*



### 3.0 FLOWCHART OF THE PROPOSED APPROACH

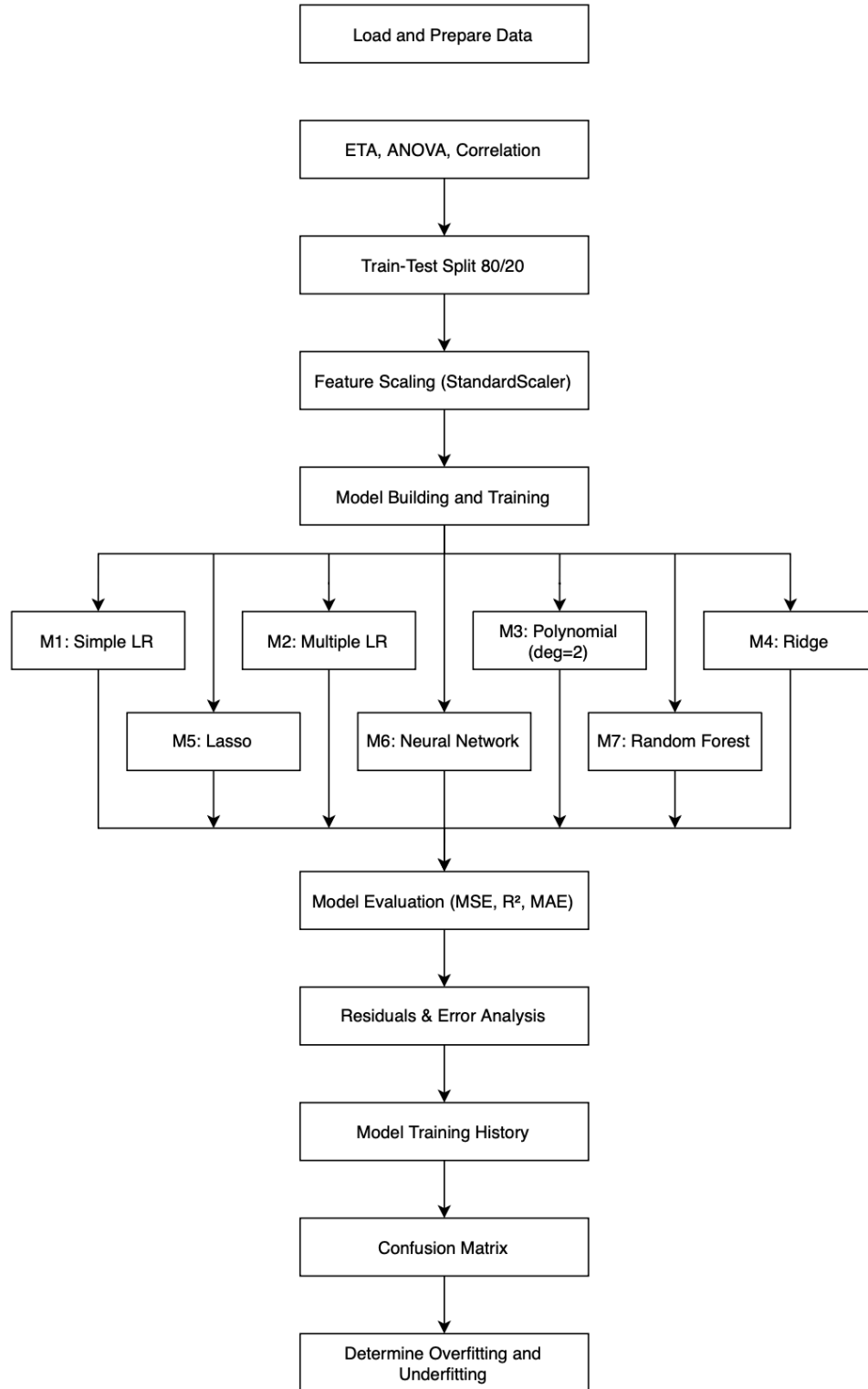


Figure 2: Flowchart of the approach

### 3.1 Exploratory Data Analysis

#### 3.1.1 Descriptive statistics

- The gene expression statistics (mean, std, min, percentiles and max) were computed across all genes and samples.

|       | AA022668   | AA045174   | AA045184   | AA054642   | AA058578   |
|-------|------------|------------|------------|------------|------------|
| count | 973.000000 | 973.000000 | 973.000000 | 973.000000 | 973.000000 |
| mean  | 0.075085   | 0.380579   | 0.329312   | 0.184106   | 0.371812   |
| std   | 0.076518   | 0.092361   | 0.186626   | 0.114658   | 0.081254   |
| min   | 0.000000   | 0.029028   | 0.024423   | 0.000000   | 0.044819   |
| 25%   | 0.024423   | 0.321276   | 0.201414   | 0.089093   | 0.316431   |
| 50%   | 0.053365   | 0.370126   | 0.257699   | 0.166126   | 0.356376   |
| 75%   | 0.098629   | 0.440007   | 0.432756   | 0.266284   | 0.419293   |
| max   | 0.558113   | 0.615230   | 0.770034   | 0.609266   | 0.608093   |

Table 5: Descriptive statistics of gene expressions (first 5 genes only)

- Detailed descriptive statistics were computed for each of the four leukemia types based on the label (only the first 5 genes are shown here).

ALL (n=436) :

|      | AA022668 | AA045174 | AA045184 | AA054642 | AA058578 |
|------|----------|----------|----------|----------|----------|
| mean | 0.095957 | 0.363487 | 0.436750 | 0.138011 | 0.350505 |
| std  | 0.097490 | 0.066484 | 0.212278 | 0.088745 | 0.054624 |
| min  | 0.000000 | 0.094814 | 0.024423 | 0.000000 | 0.110098 |
| max  | 0.558113 | 0.584845 | 0.770034 | 0.520022 | 0.539214 |

Table 6: Descriptive statistics of gene expressions for ALL samples

AML (n=257) :

|      | AA022668 | AA045174 | AA045184 | AA054642 | AA058578 |
|------|----------|----------|----------|----------|----------|
| mean | 0.058559 | 0.324695 | 0.267990 | 0.188743 | 0.323175 |
| std  | 0.048464 | 0.079834 | 0.120176 | 0.131798 | 0.059026 |
| min  | 0.000000 | 0.029028 | 0.040988 | 0.009505 | 0.044819 |
| max  | 0.295888 | 0.535772 | 0.728942 | 0.609266 | 0.524893 |

Table 7: Descriptive statistics of gene expressions for AML samples

CLL (n=237) :

|      | AA022668 | AA045174 | AA045184 | AA054642 | AA058578 |
|------|----------|----------|----------|----------|----------|
| mean | 0.053838 | 0.484737 | 0.216842 | 0.245254 | 0.472142 |
| std  | 0.045128 | 0.063525 | 0.058821 | 0.098931 | 0.061662 |
| min  | 0.000000 | 0.299248 | 0.039981 | 0.003558 | 0.303908 |
| max  | 0.221395 | 0.615230 | 0.423914 | 0.448051 | 0.608093 |

*Table 8: Descriptive statistics of gene expressions for CLL samples*

CML (n=43)

|      | AA022668 | AA045174 | AA045184 | AA054642 | AA058578 |
|------|----------|----------|----------|----------|----------|
| mean | 0.079326 | 0.313815 | 0.226332 | 0.286756 | 0.325575 |
| std  | 0.050441 | 0.046864 | 0.082445 | 0.087899 | 0.033355 |
| min  | 0.007460 | 0.186818 | 0.053202 | 0.110098 | 0.246972 |
| max  | 0.206351 | 0.419293 | 0.562838 | 0.474196 | 0.393333 |

*Table 9: Descriptive statistics of gene expressions for CML samples*

We can observe here that ALL has the largest sample size, while AML and CLL have almost the same size, and CLL has the smallest sample size.

### 3.1.2 Basic of grouping

Mean gene expression values were calculated for each gene, grouped by the 4 leukemia types showing the average expression for each leukemia class.

| Leukemia_Type | AA022668 | AA045174 | AA045184 | AA054642 | AA058578 |
|---------------|----------|----------|----------|----------|----------|
| ALL           | 0.095957 | 0.363487 | 0.436750 | 0.138011 | 0.350505 |
| AML           | 0.058559 | 0.324695 | 0.267990 | 0.188743 | 0.323175 |
| CLL           | 0.053838 | 0.484737 | 0.216842 | 0.245254 | 0.472142 |
| CML           | 0.079326 | 0.313815 | 0.226332 | 0.286756 | 0.325575 |

*Table 10: Mean gene expression grouped by leukemia types*

### 3.1.3 ANOVA

- One-way ANOVA tests were performed for each gene to identify significant differences in expression across leukemia types.
- The `anova_df` DataFrame was created, listing F-statistics and p-values for each gene.

- It was confirmed that 1378 genes (out of all 2420 genes) showed significant differences ( $p < 0.05$ ) across leukemia types.
- The top 5 genes with the most and least significant differences were displayed.

|      | Gene      | F-Statistic | P-Value  |
|------|-----------|-------------|----------|
| 8    | BG528420  | 1373.353380 | 0.000000 |
| 370  | AK025578  | 1579.511658 | 0.000000 |
| 1191 | NM_014736 | 1587.628286 | 0.000000 |
| 803  | M11722    | 1252.224851 | 0.000000 |
| 911  | NM_001071 | 1845.146364 | 0.000000 |
| ...  | ...       | ...         | ...      |
| 448  | AL568652  | 0.455956    | 0.713145 |
| 557  | AW950865  | 0.375386    | 0.770781 |
| 251  | AI539425  | 0.332968    | 0.801517 |
| 129  | AF071542  | 0.268357    | 0.848234 |
| 189  | AF427619  | 0.182052    | 0.908612 |

*Table 11: Top 5 genes with the most and least significant p-value*

### 3.1.4 Correlation

- Correlation matrices were calculated for each leukemia type.
- Summary statistics of correlation strength for each leukemia type are shown below:

| Leukemia_Type | Mean Correlation Strength ( $ r $ ) | Max Positive Correlation ( $r$ ) | Max Negative Correlation ( $r$ ) | Genes with $ r  \geq 0.7$ |
|---------------|-------------------------------------|----------------------------------|----------------------------------|---------------------------|
| ALL           | 0.1284                              | 0.9944                           | -0.8533                          | 2298                      |
| AML           | 0.1111                              | 0.9938                           | -0.7303                          | 850                       |
| CLL           | 0.1245                              | 0.9927                           | -0.7474                          | 1154                      |
| CML           | 0.1691                              | 0.9948                           | -0.8714                          | 2171                      |

*Table 12: Gene expression correlation statistics for each leukemia type*

- Genes with strong correlations (absolute correlation coefficient  $|r| \geq 0.7$ ) were identified for each leukemia type. However, for visualization clarity, only the top 25 of these strongly correlated genes were selected for each type for the heatmap visualization.

### Gene Correlation Analysis by Leukemia Type (Selected Genes with Strong Correlations)

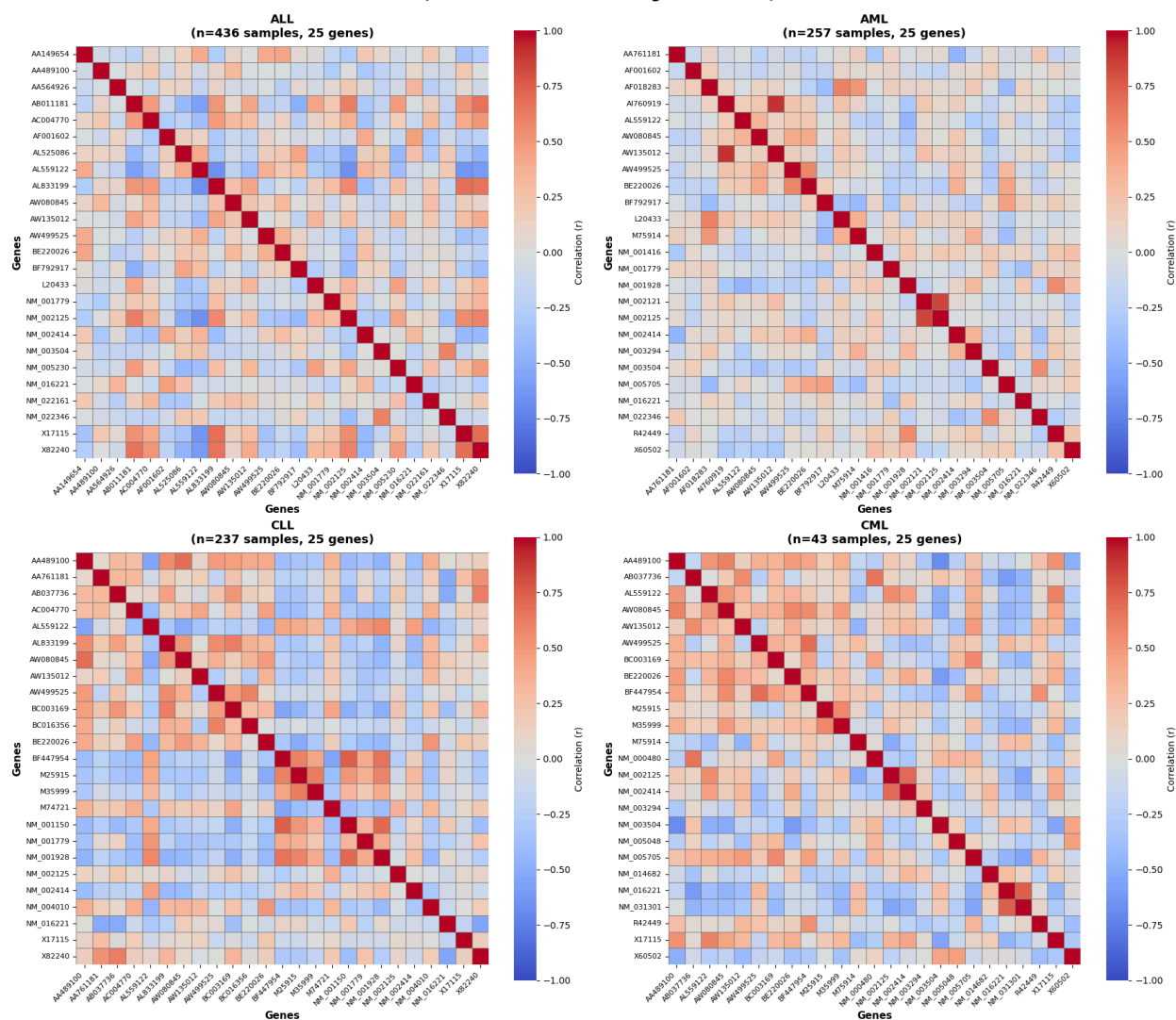


Figure 3: Heatmap of top 25 strongest correlation gene pairs for each leukemia type

### 3.2 Model Development

Before conducting the model training and testing, the dataset is split into an 80% training set and a 20% testing set. The features ( $X_{train}$ ,  $X_{test}$ ) are standardized using `StandardScaler` to ensure all genes are on a similar scale. Instead of using the leukemia type label, the target code is used such that ALL = 0, AML = 1, CLL = 2, and CML = 3.

Before analysing the effectiveness of each model, below are the MSE and R-squared of each models:

| Model              | Train MSE    | Test MSE | Train R <sup>2</sup> | Test R <sup>2</sup> | Accuracy (%) |
|--------------------|--------------|----------|----------------------|---------------------|--------------|
| Simple LR          | 8.176793e-01 | 0.833054 | 0.042449             | 0.030280            | 30.26        |
| Multiple LR        | 1.730419e-29 | 0.153705 | 1.000000             | 0.821079            | 84.62        |
| Polynomial (deg=2) | 4.880883e-28 | 0.032354 | 1.000000             | 0.962338            | 97.44        |
| Ridge              | 8.942474e-03 | 0.067925 | 0.989528             | 0.920932            | 96.41        |
| Lasso              | 4.422814e-02 | 0.059842 | 0.948206             | 0.930341            | 93.85        |
| Neural Network     | 1.250577e-02 | 0.017728 | 0.985355             | 0.979364            | 98.46        |
| Random Forest      | 9.576060e-03 | 0.061498 | 0.988786             | 0.928413            | 93.85        |

Table 13: Train and test R-squared and MSE, and accuracy for all models

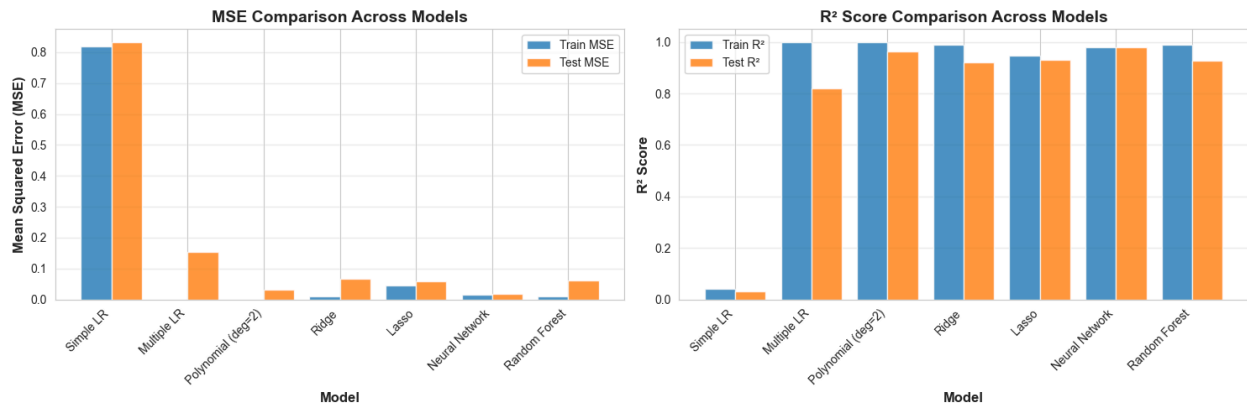


Figure 4: Train and test R-squared and MSE for all models

Below are the 7 models conducted, the results are visualized in 2 graphs: the **Actual vs Predicted results of Test Set** which visualize the model performance on the test dataset, and the **Residuals Analysis** showing the predicted result (target code) and the error (MAE).

### 3.2.1 Simple Linear Regression

A basic linear regression model is trained using only the first gene feature as a simple baseline. It achieves a low  $R^2$  score, indicating that this model **cannot effectively predict** leukemia type.

| Model     | Train MSE    | Test MSE | Train $R^2$ | Test $R^2$ | Accuracy (%) |
|-----------|--------------|----------|-------------|------------|--------------|
| Simple LR | 8.176793e-01 | 0.833054 | 0.042449    | 0.030280   | 30.26        |

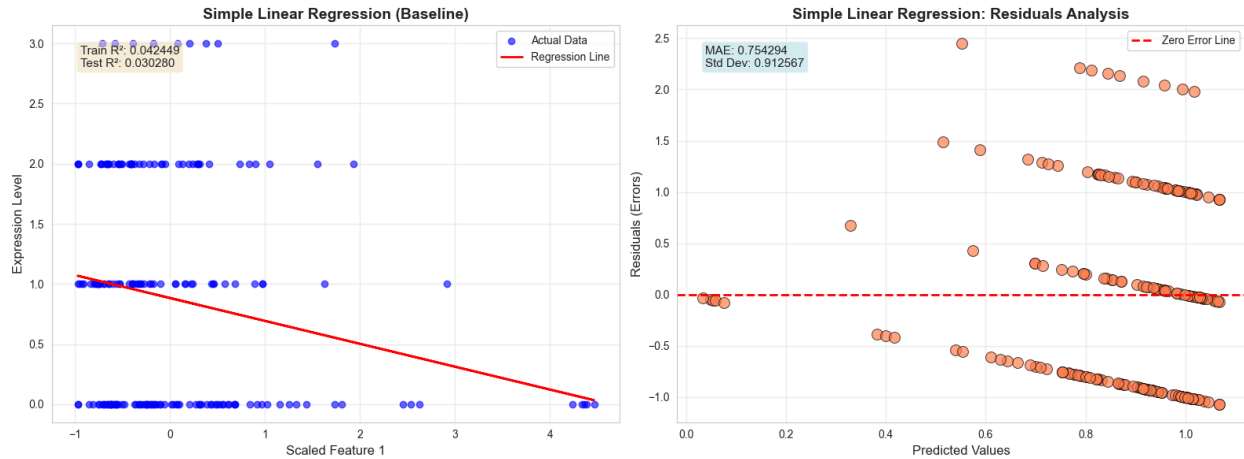


Figure 5: Simple Linear Regression Actual vs Predicted results of Test Set; Residuals Analysis

### 3.2.2 Multiple Linear Regression

This model uses all gene features to train a linear regression model. It shows a perfect  $R$ -squared of 1.0 on the training data but a negative  $R$ -squared on the test data indicating **extreme overfitting**.

| Model       | Train MSE    | Test MSE | Train $R^2$ | Test $R^2$ | Accuracy (%) |
|-------------|--------------|----------|-------------|------------|--------------|
| Multiple LR | 1.730419e-29 | 0.153705 | 1.000000    | 0.821079   | 84.62        |

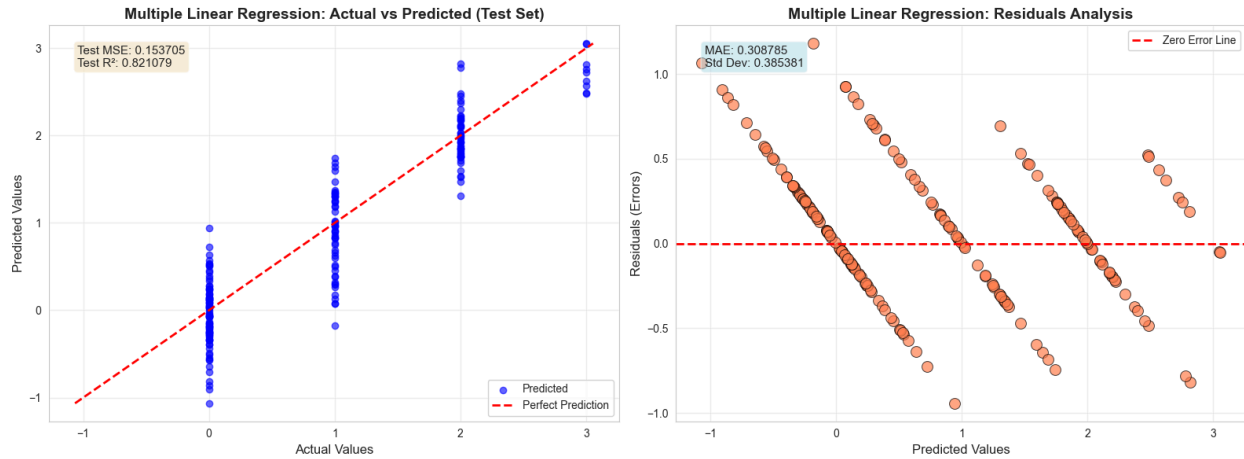


Figure 6: Multiple Linear Regression Actual vs Predicted results of Test Set; Residuals Analysis

### 3.2.3 Polynomial Regression and Pipelines

A pipeline is used to create a second-degree polynomial regression model. Similar to the multiple linear regression model, it severely overfits the training data, resulting in perfect training metrics but very poor (negative  $R^2$ ) performance on the test set.

| Model              | Train MSE    | Test MSE | Train R² | Test R²  | Accuracy (%) |
|--------------------|--------------|----------|----------|----------|--------------|
| Polynomial (deg=2) | 4.880883e-28 | 0.032354 | 1.000000 | 0.962338 | 97.44        |

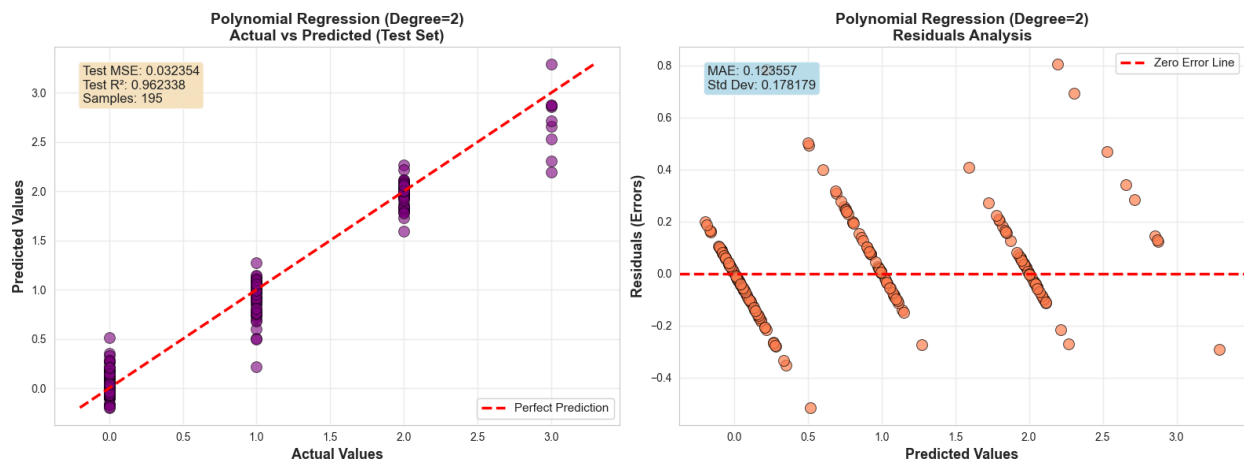


Figure 7: Polynomial Regression Actual vs Predicted results of Test Set; Residuals Analysis



### 3.2.4 Ridge Regression (L2 Regularization)

Using GridSearchCV, the best regularization strength ( $\alpha$ ) is found to be 100. This model shows much better generalization, with a test  $R^2$  score of approximately 0.81.

| Model | Train MSE    | Test MSE | Train $R^2$ | Test $R^2$ | Accuracy (%) |
|-------|--------------|----------|-------------|------------|--------------|
| Ridge | 8.942474e-03 | 0.067925 | 0.989528    | 0.920932   | 96.41        |

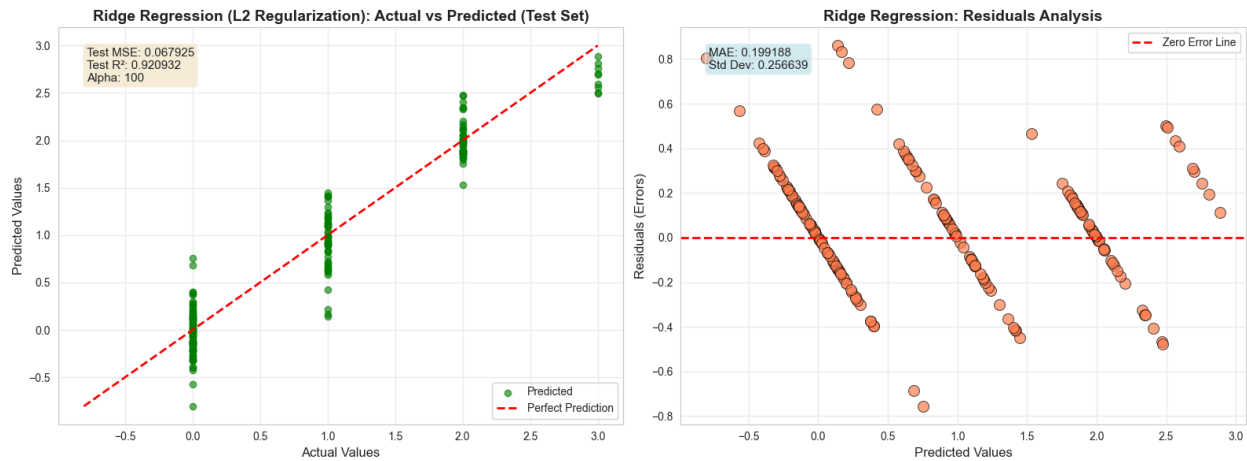


Figure 8: Ridge Regression Actual vs Predicted results of Test Set; Residuals Analysis

### 3.2.5 Lasso Regression (L1 Regularization)

Lasso regression is used for both regularization and feature selection. It identifies the best  $\alpha$  as 0.1 and reduces the number of features from 1422 to just 20. It achieves a test  $R^2$  score of about 0.83, slightly outperforming Ridge.

| Model | Train MSE    | Test MSE | Train $R^2$ | Test $R^2$ | Accuracy (%) |
|-------|--------------|----------|-------------|------------|--------------|
| Lasso | 4.422814e-02 | 0.059842 | 0.948206    | 0.930341   | 93.85        |

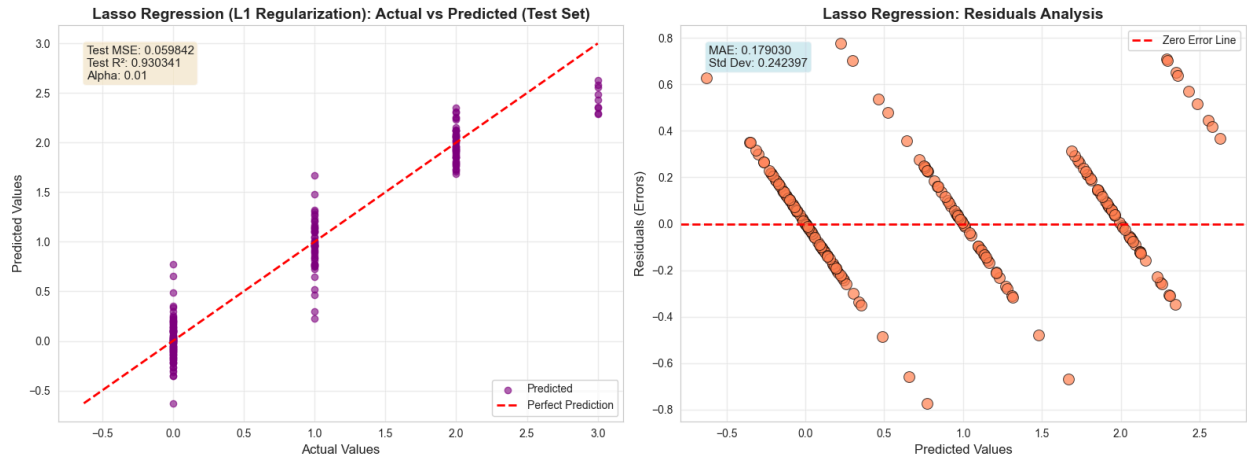


Figure 9: Lasso Regression Actual vs Predicted results of Test Set; Residuals Analysis

### 3.2.6 Neural Network

A deep neural network is built using Keras, featuring multiple dense layers with `relu` activation, `BatchNormalization`, and `Dropout` to prevent overfitting. It is trained with an `Adam` optimizer and `EarlyStopping`. This model achieves the highest test R<sup>2</sup> score of approximately 0.90.

| Model          | Train MSE    | Test MSE | Train R <sup>2</sup> | Test R <sup>2</sup> | Accuracy (%) |
|----------------|--------------|----------|----------------------|---------------------|--------------|
| Neural Network | 1.250577e-02 | 0.017728 | 0.985355             | 0.979364            | 98.46        |

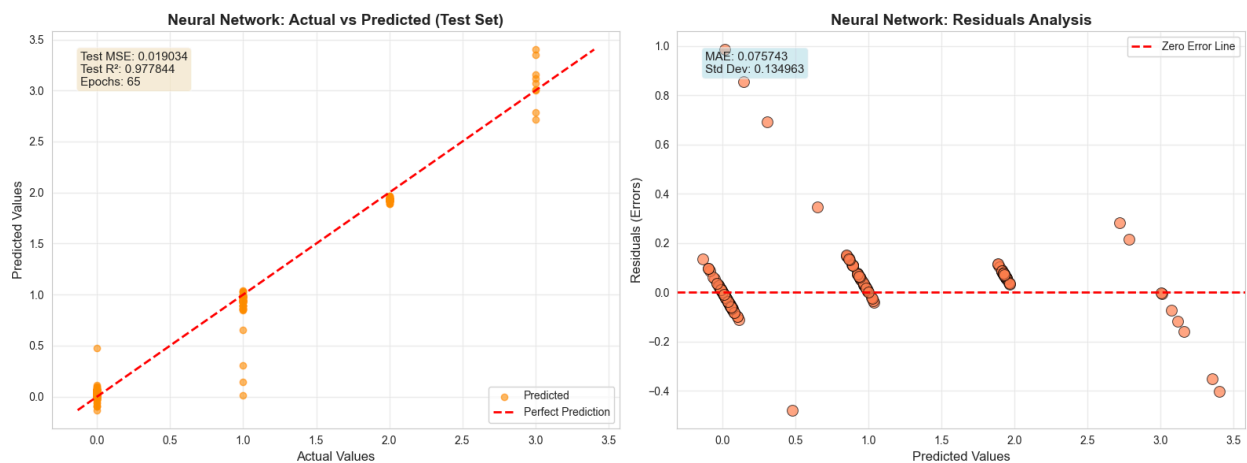


Figure 10: Neural Network Actual vs Predicted results of Test Set; Residuals Analysis

### 3.2.7 Random Forest Regressor

A Random Forest model is trained, and GridSearchCV is used to tune its hyperparameters. It achieves a very high training  $R^2$  but a lower test  $R^2$  of around 0.82, showing some overfitting but still performing well.

| Model         | Train MSE    | Test MSE | Train $R^2$ | Test $R^2$ | Accuracy (%) |
|---------------|--------------|----------|-------------|------------|--------------|
| Random Forest | 9.576060e-03 | 0.061498 | 0.988786    | 0.928413   | 93.85        |

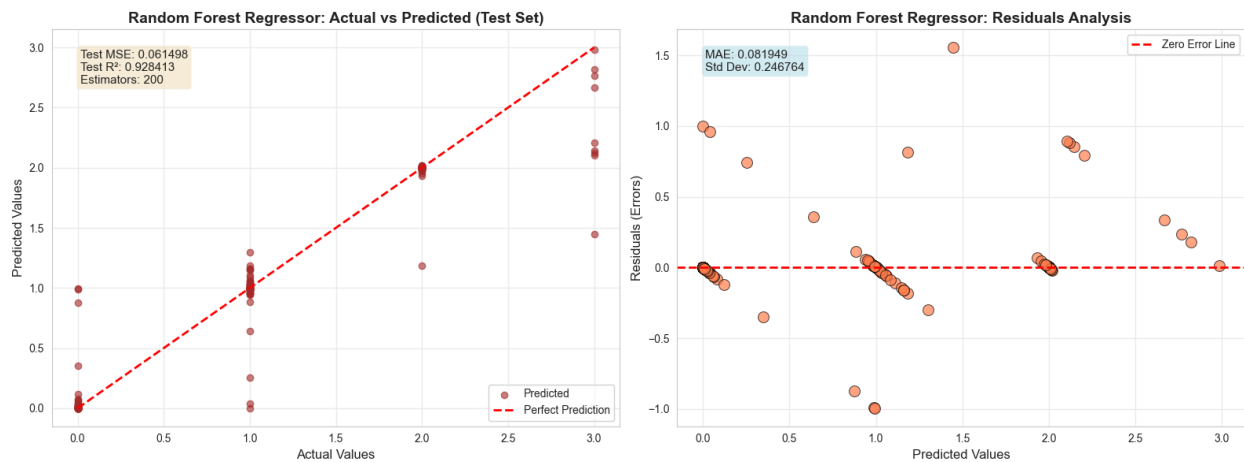


Figure 11: Random Forest Actual vs Predicted results of Test Set; Residuals Analysis

### 3.2.8 Prediction And Decision Making

The Neural Network is identified as the best-performing model based on the highest test  $R^2$  and lowest test MSE among all 7 models. The overall classification accuracy of the NN model is 98.46, with 192 correct diagnosis out of all 195 test set samples. The details look into the testing performance of NN are as follows:

| Leukemia_Type | Accuracy_Percentage | Test_Set_Ratio |
|---------------|---------------------|----------------|
| ALL           | 100.00              | 87 / 87        |
| AML           | 94.23               | 49 / 52        |
| CLL           | 100.00              | 47 / 47        |
| CML           | 100.00              | 9 / 9          |

Table 14: Accuracy of the Neural Network model in predicting each leukemia type

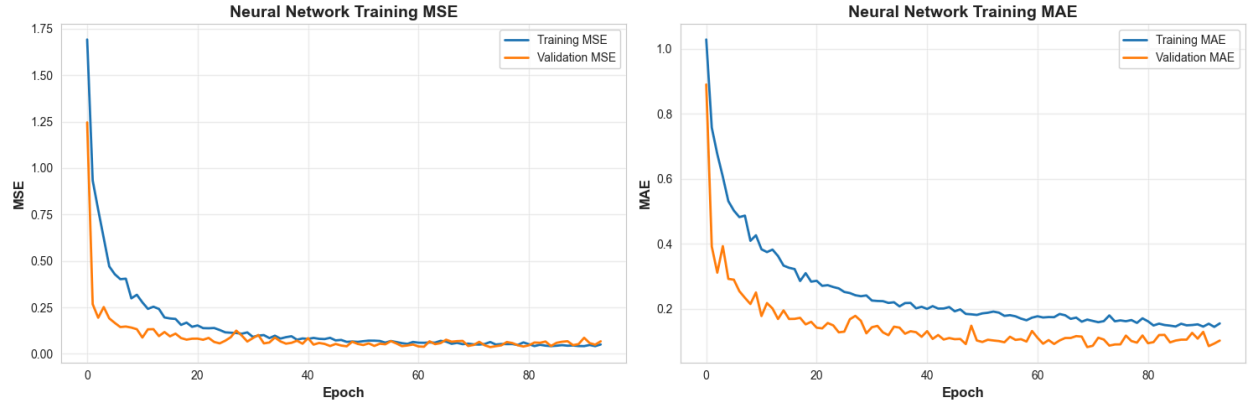
| Sample_ID | Actual_Type | Predicted_Type | Prediction_Error |
|-----------|-------------|----------------|------------------|
| GSM331892 | AML         | ALL            | 0.985746         |
| GSM332549 | AML         | ALL            | 0.985746         |
| GSM332761 | AML         | ALL            | 0.985746         |

*Table 15: Misclassified test set samples of the Neural Network model*

### 3.3 Model Evaluation

#### 3.3.1 Neural Network Training History

To better understand the training and testing of Neural Network model, its training history is visualized to show the training and validation Mean Square Error (MSE) and Mean Absolute Error (MAE) curves across the epochs. This is used to assess how the model learned over time and to confirm that it converged properly without excessive overfitting during training.



*Figure 12: Training History of Neural Network Model*

### 3.3.2 Confusion Matrix



Figure 13: Confusion Matrix of all models

The ideal model should show a confusion matrix with 100% along the diagonal axis. While in reality the Neural Network model is good enough in showing only 3 false negative ‘AML’, with other 3 leukemia types scoring 100% accuracy. It is interesting to note that 4 out of 7 models perfectly predicted the ‘CLL’ type, potentially due to the existence of a significant biomarker. While no model can perfectly predict the ‘AML’ type. This insight is similar to the correlation heatmap where CLL has stronger correlations, while AML has weaker correlation among gene pairs, making AML more difficult to be detected and predicted accurately..

### 3.3.3 Over-fitting and Under-fitting

This part compares the training versus testing  $R^2$  and MSE scores for all developed models. By analyzing the gap between training and test performance, a large difference between training and testing performance (high  $R2\_Diff$  or  $MSE\_Diff$ ) indicates overfitting, where the model has learned the training data too well, including its noise, and fails to generalize to new, unseen data. Conversely, poor performance on both training and testing sets suggests underfitting.

| Model              | $R2\_Diff$<br>(Train - Test) | $MSE\_Diff$<br>(Test - Train) | Underfitting/Overfitting                                                                                                                                                         |
|--------------------|------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simple LR          | 0.012170                     | 0.015375                      | No significant underfitting/overfitting, because both train and test data show low R-squared and high MSE scores, indicating that the model is too simple for this complex data. |
| Multiple LR        | 0.178921                     | 0.153705                      | Highly overfitting, showing biggest gap between train and test scores among all model.,                                                                                          |
| Polynomial (deg=2) | 0.037662                     | 0.032354                      | Overfitting                                                                                                                                                                      |
| Ridge              | 0.068596                     | 0.058982                      | Overfitting                                                                                                                                                                      |
| Lasso              | 0.017865                     | 0.015613                      | Good Fit, but due to lower R-squared in comparison with other models, it is not an accurate model.                                                                               |
| Neural Network     | 0.005991                     | 0.005222                      | Best Generalization                                                                                                                                                              |
| Random Forest      | 0.060373                     | 0.051922                      | Overfitting                                                                                                                                                                      |

*Table 16: Misclassified test set samples of the Neural Network model*

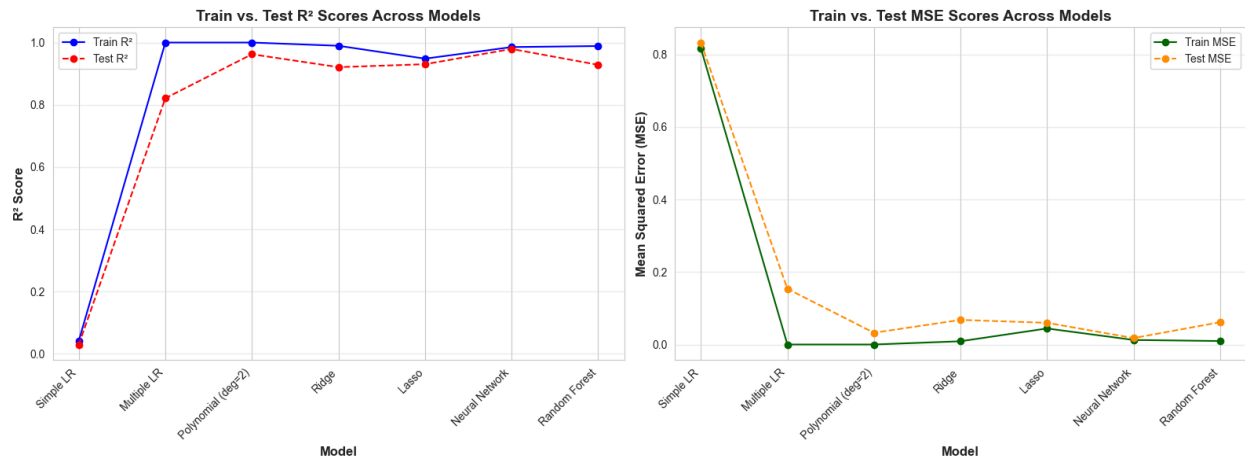


Figure 14: Train vs Test R-squared and MSE score across models

## 4.0 CONCLUSIONS

This project has successfully developed and evaluated 7 machine learning models to classify 4 major types of leukemia—ALL, AML, CLL, and CML, using gene expression data from the GEO dataset GSE13164. The initial phase involved a data wrangling pipeline to download, clean, and process the raw microarray data into the csv format. This included mapping thousands of probes to their corresponding genes and aligning the expression data with sample labels, resulting in a high-dimensional dataset with over 1,400 gene features for over 900 samples. Exploratory Data Analysis, including ANOVA testing and correlation analysis, confirmed that significant statistical differences in gene expression exist among the leukemia types, validating the dataset's potential for building a classifier.

A total of 7 models were trained and tested to tackle this classification problem using 80/20 split, and the 4 classes are encoded into numerical targets (0-3) instead of categorical. The initial baseline which is the Simple LR model was too basic to capture any meaningful patterns, while the Multiple Regression models memorized the training data but failed on the test set showing the biggest difference between train and test performance. This highlighted the critical need for regularization to handle the high-dimensional nature of the data. The introduction of regularization, which are Ridge, Lasso, and Random Forest models, proved to be effective, achieving high accuracy and demonstrating better generalization from training to test data. However, the deep Neural Network outperforms other models, achieving the highest test accuracy of 98.46%.

In conclusion, this project demonstrates that machine learning, particularly deep learning, can be a powerful tool for leukemia diagnosis from gene expression data. This gives insights into bioinformatics approaches in cancer diagnosis that may extend beyond the scope of leukemia diagnosis. On another hand, it would be interesting to apply this approach on the sample obtained from leukemia patients over their chemotherapy sessions and even after they are cancer-free, to better predict potential leukemia recurrences.



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